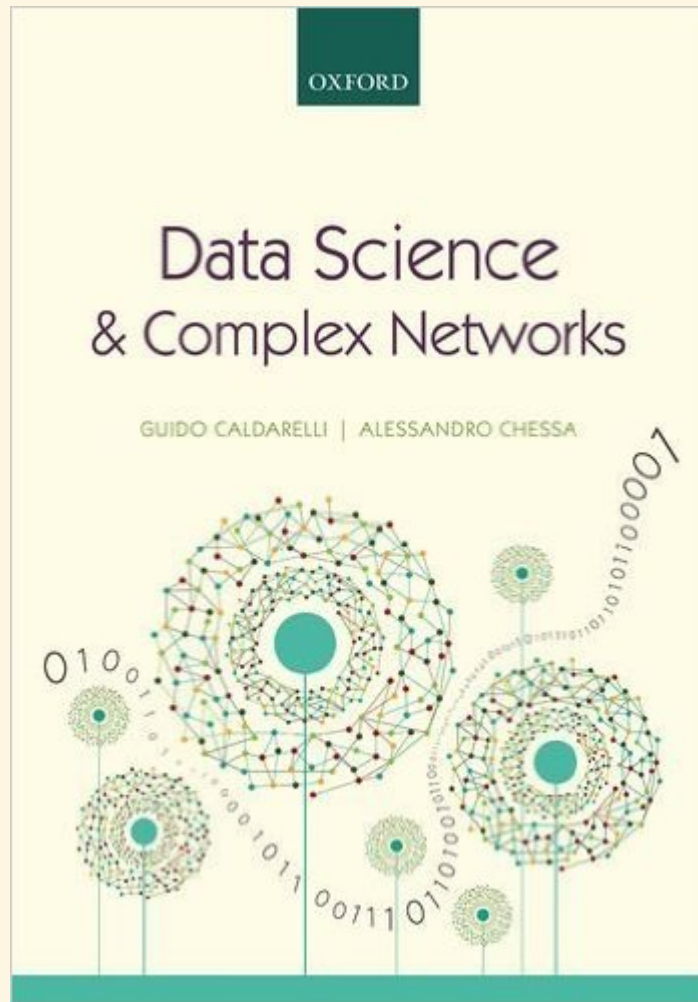


# FOOD WEBS

AP

# DATA SCIENCE AND COMPLEX NETWORKS



# BASIC CONCEPTS

The study of how objects(entities) connect to each other and the properties of their connection.

Possible understanding: a relationship instance.

# TERMINOLOGY

- $G = \langle V, E \rangle$  where  $E \subseteq V \times V$
- $|V| = n, |E| = m$ , density is  $|E|/|V|^2$
- vertex  $v$  is adjacent to  $u$  if  $(u, v) \in E$ ;  $(v, v) \notin E$ .
- neigh. of  $v$ ,  $N(v)$ : the set of adjacent vertices;  $\deg(v) = |N(v)|$
- The adjacency matrix  $A_{n \times n}$  of  $G$ :  $(u, v) \in E \leftrightarrow a_{ij} = 1$
- [incidence matrix  $I_{n \times m}$  of  $G$ :]

# PATHS, CONNECTEDNESS

- A path  $u \rightarrow v$  is a *sequence* of edges  $\langle (u, c_1), (c_1, c_2), \dots, (c_k, v) \rangle$
- its length  $(k+1)$  is the cardinality of the path.
- Two vertices are connected if  $\exists$  a path betw. them.
- A graph is connected if all its vertices are.

# DISTANCES, I

- Distance is the length of the (possibly non-unique) shortest path connecting them,  $\infty$  otherwise.
- The diameter of a graph is the maximum distance between any two vertices

## DISTANCES, II

Average distances are also important

[...] the first world-scale social-network graph-distance computations, using the entire Facebook network of active users (~721 million users, ~69 billion friendship links). The average distance [...] is 4.74, corresponding to 3.74 intermediaries or “degrees of separation.”

# WEIGHTED, DIRECTED, MULTIPLEX

$G = \langle V, E, w \rangle$  where  $w : E \rightarrow \mathbb{R}$

Path length: sum of the weights of the arcs.

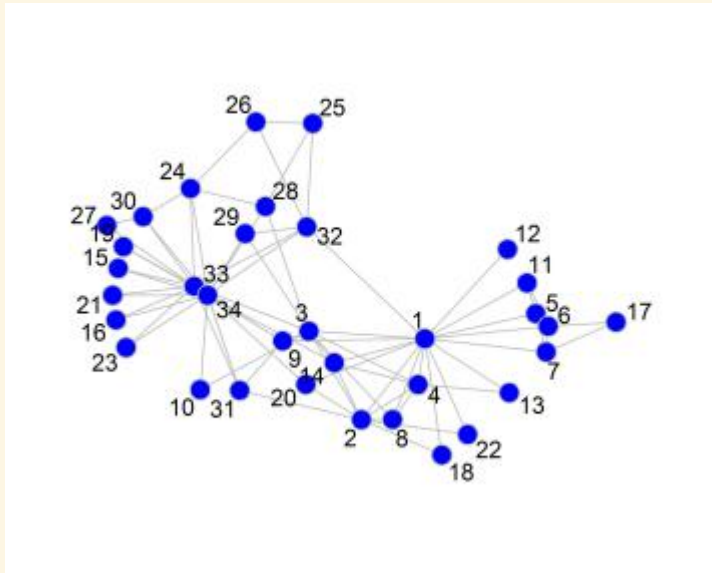
$G = \langle V, E \rangle$  where  $v \in V$  are **nodes** and  $\langle u, v \rangle \in E$  are **arcs**.  $w : E \rightarrow \mathbb{R}$

Out-neigh. and In-neigh.

$G = \langle V, E, D \rangle$  where  $V$  are **nodes**,  $D$  are **dimensions/layers** and  $\langle u, v, d \rangle \in E$  are **arcs**

# INTERESTING QUESTIONS

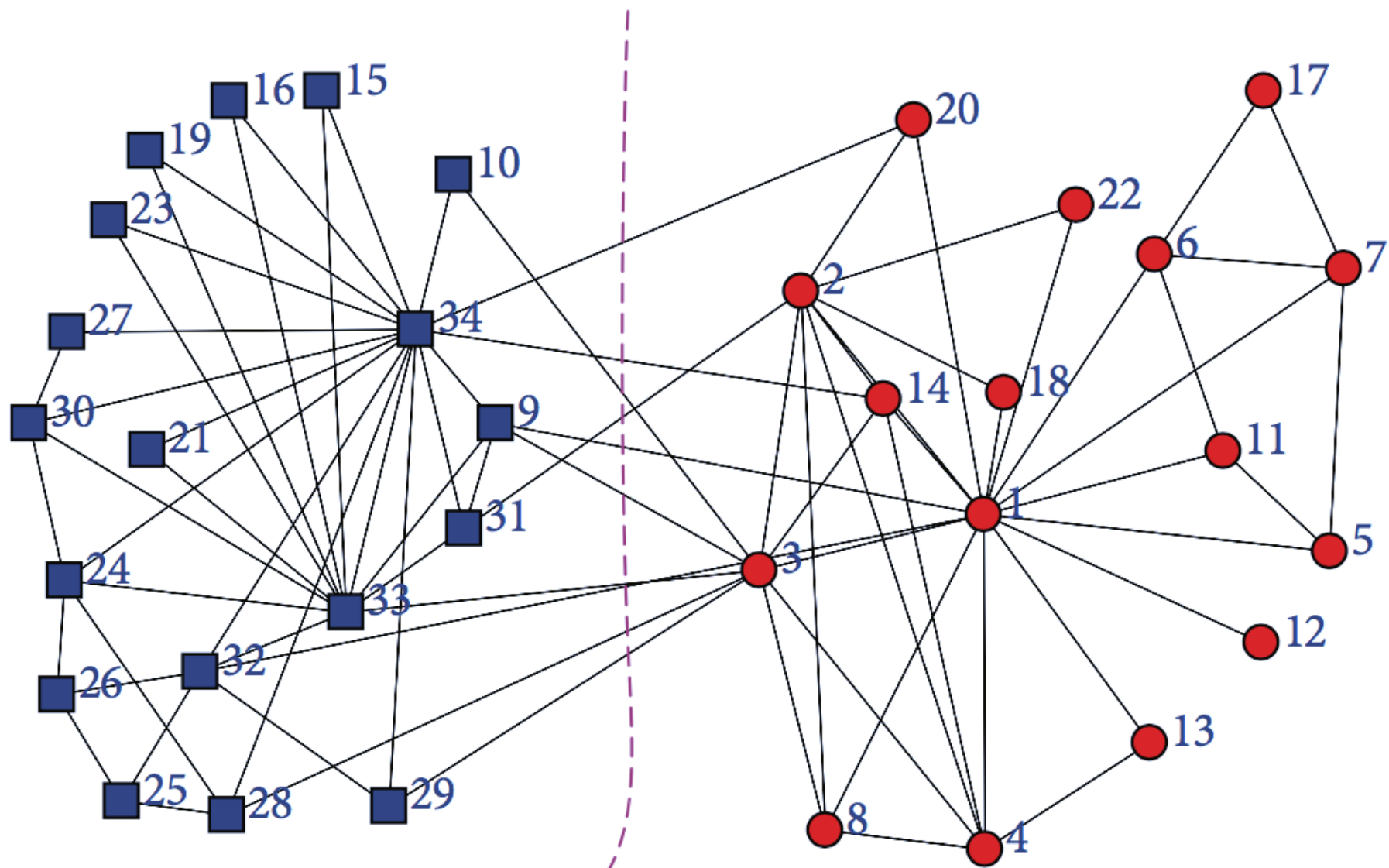
# INTERESTING QUESTIONS I: PATHS AND *FLOW*



## INTERESTING QUESTIONS II: CENTRALITIES



# INTERESTING QUESTIONS III: CLUSTERING

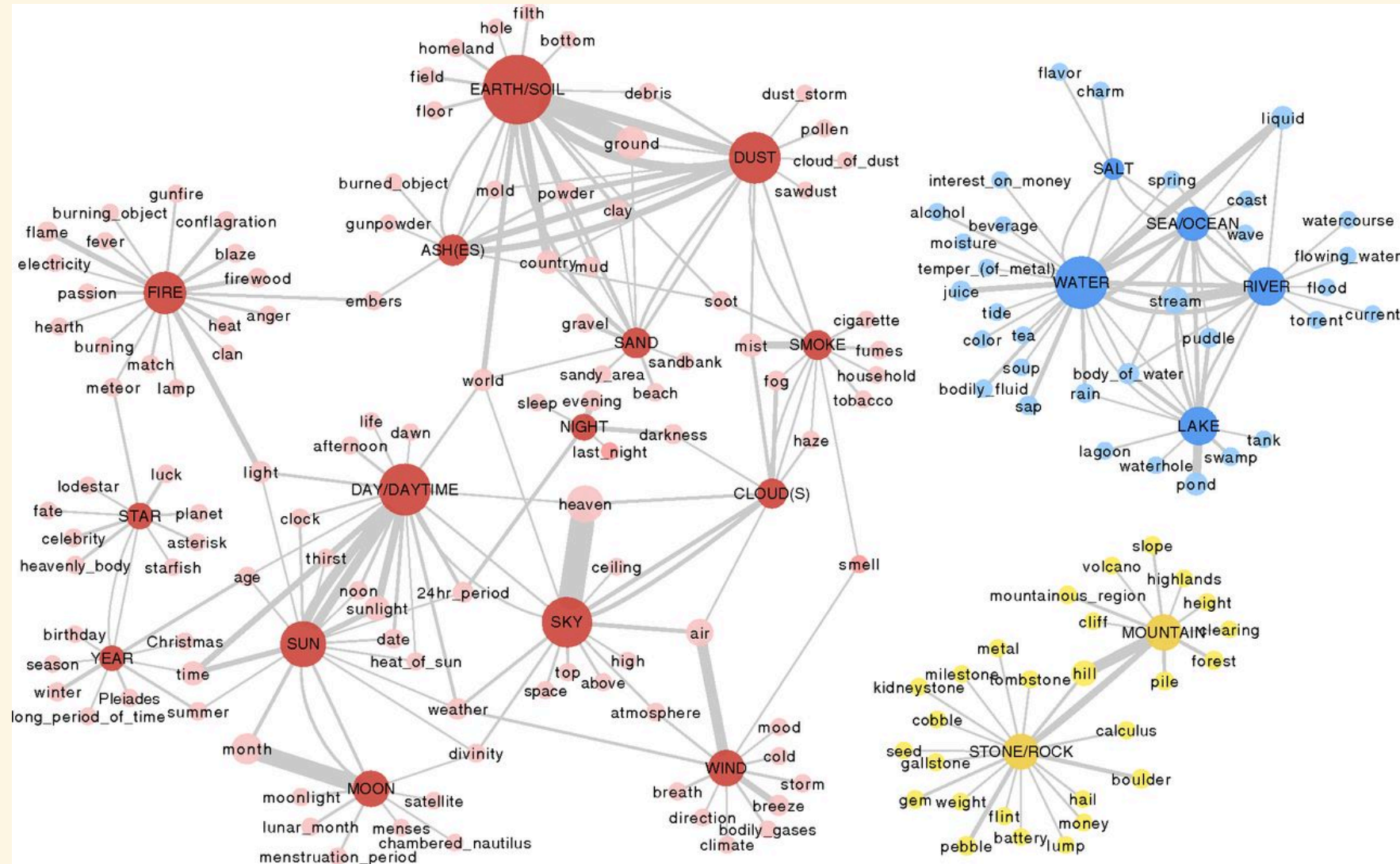


The CGC clustering

```
1 import networkx as nx
2
3 G = nx.karate_club_graph()
4
5 print("Node Degree")
6
7 for v in G:
8     print('%s %s' % (v, G.degree(v)))
```

# EXTRA: VISUALIZING FREQUENCIES BY GRAPHS

## Polysemy:

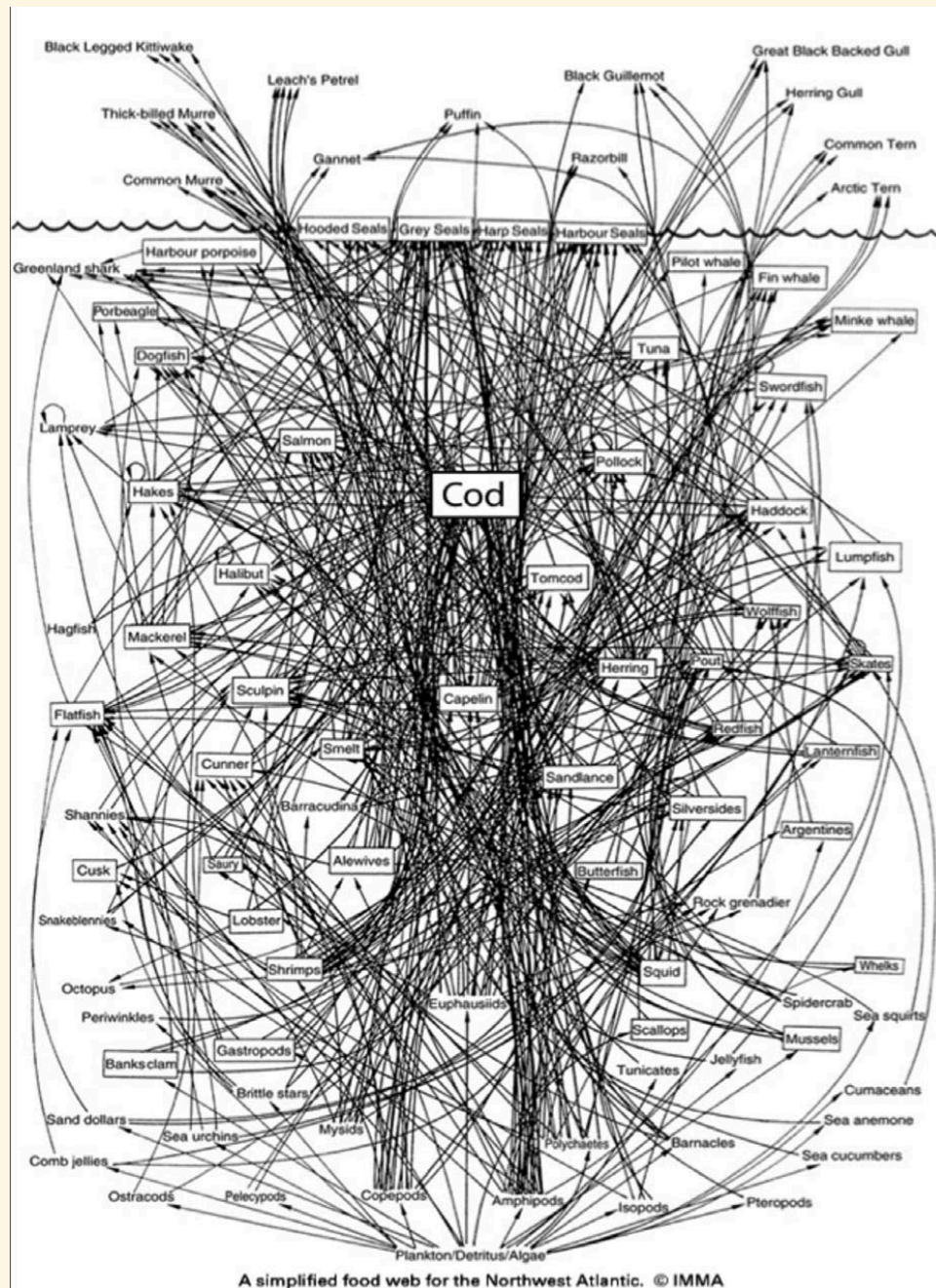


# CH. 1: FOOD WEBS

# IMPORTANT CONCEPTS

- load data and organise it in a **networkx** data structure
- modeling tip: it is ok to have a special node representing “nature”
- modeling tip: look for *invariants*
- find the connected component (the **bowtie**):

source, connect and sink.



# LEARN NETWORK SCIENCE

# METHOD

- study *degree distribution*
- find properties of a network in terms of the *degree organization*
- study *clustering coefficient*: why is it so much better than plain network density?

Clone the [original \(Python 2\)](#) from Github

From the same author, a [summary of the main concepts](#)

